

Exercise 1: Querying the Key Management System

A KMS (Key Management System) stores and manages keys created by QKD service. One established protocol to communicate with a KMS is the ETSI 014 standard.

Two KMS are running on `http://tii-qkd-workshop.uaenorth.cloudapp.azure.com`

- "Master" SAE ID `K002X001` is running on port 5001
- "Slave" SAE ID `K002X002` is running on port 5002

TASK: Develop a key retrieval client to interact with the KMS and implement the following API calls.

No.	Method name	URL	Access method
1	Get status	<code>http://{KME_hostname}/api/v1/keys/{slave_SAE_ID}/status</code>	GET
2	Get key	<code>http://{KME_hostname}/api/v1/keys/{slave_SAE_ID}/enc_keys</code>	POST (or GET)
3	Get key with key IDs	<code>http://{KME_hostname}/api/v1/keys/{master_SAE_ID}/dec_keys</code>	POST (or GET)

Subtasks

To complete the task, be able to answer and complete the following subtasks:

1. How many keys are in the KMS?
2. Retrieve any key in JSON format
3. Retrieve key with ID `f7e0edef-8bf4-43b2-983b-1e9d22ca6f13`